

FU DAI
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Work Background

Over 10 years of cross-industry experience of cross-industry experience in data science, intelligent manufacturing. Led multiple cross-functional projects. Proven track record in building production-grade AI systems, integrating domain knowledge with scalable MLOps practices. Certified Huawei HCIE Big Data Expert and PMP Project Manager. Author of peer-reviewed petroleum engineering research and inventor of AI patents.

TECHNICAL SKILLS

- Machine Learning & AI: Scikit-learn, XGBoost, PyTorch, LangChain, WhisperX, Federated Learning
- MLOps & Cloud: Kubernetes, Helm, Docker, Jenkins, Git, MLflow, SwanLab, vLLM, LLMDeploy, Nginx
- Big Data & Pipeline: Spark, Hive, Kafka, HDFS
- DB & Storage: Milvus, ClickHouse, ElasticSearch, MySQL
- Computer Vision: OpenCV, TensorRT, OpenVINO
- Security & FL: MPC, RSA, Hash encryption protocols
- Programming & Scripting: Python, SQL, Shell

Work Experience

Oct 2023 – Present **G42 Presight, UAE** **Data Scientist**

Company intro: AI and big data company serving government and critical sectors in the UAE.

- Facial Recognition Application Development: Model training, quantization for lightweight and mobile deployment. Large-scale test (~6 million identities), algorithm improvement (backbone replacement: iresnet-> vit, mabavision, flash attention module), improved accuracy in complex scenarios such as low light, and occlusions, wearing glasses.
- RAG System deployment & modification: Deployed Ragflow using K8s Helm; integrated Suya OCR for Arabic document recognition; Supported LLM chatbot for document Q&A and served emergency/Gov Q&A use cases.
- Voice bot customization: Integrate WhisperX+LLM+TTS_v2 for supporting English/Arabic/Hindi. Modify transfer protocol to support local customization, supported speech subtitle generation and real-time semantic interaction.
- Public Opinion Analysis and Multi-source Fusion System: integrated multiple data sources (GDELT, self-collected Middle East news sources), built ETL and LLM-based NLP pipeline for news labeling, translation and aggregation. Integrated with ElasticSearch and ClickHouse to support real-time keyword monitoring.
- Model/Data Management: VLLM deployment; built MLFlow and SwanLab to manage ML metric indicators.
- Kubernetes and Jenkins Automated Deployment: Completed semi-automated CI/CD process from git commit → Jenkins compile → Helm upgrade. Managed multiple services with Supervisor to ensure stable operation of ETL and NLP pipeline services.

Oct 2021 – Oct 2023 **Xiaomi Mobile Software Co., Ltd., Beijing** **Senior Algorithm Engineer**

Company intro: subsidiary of Xiaomi Group, specializing in IOT software and intelligent manufacturing system development.

- Led APS (Advanced Planning and Scheduling) System Design: system prototyping, built rule-based scheduling engine. Explored the application of reinforcement learning and operations research algorithms (e.g., integer programming, heuristic search) in multi-objective scheduling scenarios (e.g., due dates, capacity, switching costs).
- Participated in National Key Project: Contributed to a Ministry of Science and Technology major initiative,

Responsible for APS solution implementation, system deployment, testing and evaluation.

- Developed Industrial Visual Defect Detection System: built an AI algorithm pipeline for component placement defect detection. Designed an end-to-end detection flow including image acquisition, region localization (OpenCV mark template matching), silk-screen OCR recognition (based on DBNet + RCNN + MobileNet), feature fusion, and defect classification. improving production line efficiency by 30%.

Apr 2019 – Sep 2020

Siemens (China) SFAE

Algorithm Engineer

Company Overview: Digital transformation division focused on industrial intelligence and data-driven manufacturing optimization solutions.

- Federated Learning Development: Designed and implemented vertical federated learning models including vertical linear regression (LR), decision trees, and CNN. Built feature importance tools such as Kernel LIME and SHAP. Completed benchmark testing and comparisons with platforms such as Tencent FATE and TFE.
- Predictive Maintenance with Vibration Signals: Used machine learning techniques for predictive maintenance. feasibility studies for production scheduling and researched operations research-based optimization strategies.

Sep 2016 – Apr 2019

OPT Petroleum Technology Co., Ltd

Data Analyst, Reservoir Engineer

Company Overview: providing high-precision reservoir modeling and production optimization support for the oil and gas industry.

- Reservoir Production Performance Analysis Software Development: Developed tools for production decline analysis and water cut forecasting. Supported quantitative decision-making for rolling development plans and dynamic production adjustment.
- Smart Diagnostic and Analysis Tools Based on Production Data: Created intelligent diagnostic tools using field production data, including dynamometer card interpretation (for beam-pumping unit condition monitoring), water cut prediction, and reservoir classification models based on petrophysical parameters and production response. Enabled intelligent well management and detailed reservoir characterization.
- Reservoir Simulator Benchmarking and Optimization: Participated in performance benchmarking and result validation between the in-house black oil simulator "Simon" and the commercial simulator Eclipse.

Aug 2012 – Aug 2014

CNOOC (China) Ltd

Reservoir Engineer

Company Overview: Engaged in offshore oil and gas exploration, development, and technical services in the South China Sea.

- Production Performance Analysis for LF3-2 Oilfield: Supported production monitoring and analysis, conducted single-well and block-level decline trend tracking and anomaly diagnostics.
- Reservoir Simulation and Optimization for LF13-2 Oilfield: numerical reservoir simulation and remaining oil distribution analysis.

Project Experience

Mar 2023 – May 2024

Facial Recognition System Development & Optimization

Project Lead

Project Overview: The goal was to maintain high recognition accuracy in complex environments (e.g., occlusion, low light) while supporting on-device deployment on Android for privacy and real-time performance.

Computer Vision Modeling: Led the development of a complete facial recognition system, covering data collection, quality evaluation (occlusion, angle, lighting), model training, deployment, and validation.

Robustness Enhancement: Fine-tuned the model using occlusion-specific datasets such as MAFA, COFW, and RealWorldOccludedFaces. Achieved ~17% improvement in recognition accuracy under conditions such as eyeglasses, masks, side faces, and low lighting, reaching 97.8% accuracy.

Model Development and Optimization: Model improvement (using Swin-ViT, MambaVision, Flash Attention backbone) to support large-scale identity cases (600M IDs). Applied parallel training, gradient accumulation, and

mixed-precision (fp16) training strategies. BatchNorm fusion, and MNN/TensorRT compilation for deployment. Achieved CPU inference latency of 150ms per request and successfully converted models from ONNX to Android AAR format.

System Integration and Evaluation: Built system modules supporting dynamic face registration and updating, embedding extraction, and Milvus vector database updates. Conducted multi-model comparison using ROC curves and TPR@FAR=1e-4 metrics.

Oct 2023 – May 2024 Large Language Model and RAG System Development Core Team Member

Project Overview: Use multiple intelligent Q&A systems based on large language models (LLMs) to enable multi-source knowledge retrieval and response.

RAG Implementation: Built a retrieval-augmented generation (RAG) framework using LangChain + Milvus. Carry out document chunking, vector generation, retrieval & re-ranking, and context concatenation to streaming output. Investigated GraphRAG to improve retrieval precision and response latency, enabling dynamic context expansion and multi-turn questioning. Supported multilingual OCR by integrating high-accuracy Suya OCR modules.

Multi-Source Information Integration: Developed pipelines for news scraping and translation; integrated sources such as GDELT and local Middle East news feeds. Implemented structured event extraction and labeling (topic/category/sentiment/entity). Processed over 300,000 news articles, achieving daily incremental ingestion and tag aggregation.

Deploy Deep Research Tools: Deployed and integrated several deep research tools.

Dec 2020 – Sep 2021 | Federated Learning Platform Development | Core R&D Member

Project Overview: Developed a vertical federated learning platform to enable cross-institutional collaborative modeling and analysis without exposing raw data, supporting heterogeneous data scenarios.

Vertical Federated Learning Algorithm Development: Built a model library for vertical federated learning, covering linear regression, multi-class logistic regression, and Poisson regression. Participated in algorithm pipeline design and was responsible for algorithm implementation and engineering optimization. Supported asynchronous updates and communication compression strategies in federated environments.

Interpretability and AutoML Module: Developed feature importance evaluation tools for federated environments, including Federated Kernel LIME and Sharply methods. Developed federated statistical operators and scorecard computation modules based on MPC protocols. Supported commonly used encryption methods.

Platform Validation and Benchmarking: Participated in standardized evaluation and performance benchmarking organized by the China Academy of Information and Communications Technology (CAICT). Benchmarked against mainstream open-source frameworks (e.g., FATE, TFE).

Apr 2020 – Sep 2020 | Intelligent Operation & Maintenance System for Shijiazhuang Steel Plant Motors (Fault Prediction) | Project Lead

Project Overview: To mitigate the risk of production line downtime caused by unexpected failures of critical motor equipment, a predictive maintenance system based on real-time data analysis was developed.

Data Processing and Feature Engineering: Used Hive + Spark SQL to extract operational data for each motor (e.g., current, voltage, temperature, vibration signals) and structured the features. For vibration signals, applied sliding window aggregation to compute statistical features (mean, standard deviation, kurtosis, skewness, RMS, frequency domain amplitude, etc.). Performed feature correlation analysis using Spark Correlation to identify relationships with fault labels and select key features.

Model Training and Evaluation: Built a training pipeline using XGBoost from Spark MLlib, completing data standardization, feature selection, model training, and validation. Hyperparameter tuning and used cross-validation. The final model achieved 87% accuracy on the test set and supported classification of multiple fault types (e.g., bearing looseness, shaft misalignment, rotor imbalance).

Business Outcome: Successfully deployed on critical production lines, covering 10+ types of equipment faults (e.g., bearing wear, looseness, abnormal vibrations). Achieved an average early warning lead time of over 2 hours, reduced the annual fault rate by over 25%, and significantly lowered maintenance costs and downtime risks.

Jul 2016 – Nov 2016 | LF13-2 Offshore Low-Permeability Reservoir Adjustment & Remaining Oil Recovery Study | Project Contributor

Numerical Simulation & Scenario Comparison (Eclipse + Tnavigtor): Developed black oil models in Eclipse and conducted history matching. Simulated the impact of injection-production patterns and development strategies (e.g., well conversion, layer adjustment, reperforation) on water cut and recovery factor. Compared with Tnavigtor simulations to evaluate the effectiveness of different strategies.

Development Plan Design & Production Forecasting: Proposed well pattern optimization and reservoir reconfiguration plans based on simulation results. Generated production forecast curves for different scenarios.

Reserves Self-Evaluation: Prepared listing reserve estimates in compliance with SEC standards based on simulation results.

Publications & Patents

➤ U.S. Patent (In Process):

Enhanced Face Recognition Training Method Based on Data Augmentation and Multi-Information Fusion

➤ China Invention Patents (Granted):

PF2201945CN – CN2022112827912: AOI Teaching Data Analysis Method, Device, Equipment, and Storage Medium

PF2206319CN – CN2022115433768: Fault Detection Method, Device, and Storage Medium

CN113254943A: Model Contribution Evaluation System Based on Vertical Federated Learning

Software Copyright: 2023SR0364976: Intelligent Production Scheduling Software for 3C Electronic Assembly Industry

➤ Journal Publications (SCI Indexed):

Dai Fu, Hadi Belhaj, Achinta Bera. Modeling and Simulation of Transition Zones in Tight Carbonate Reservoirs by Incorporation of Improved Rock Typing and Hysteresis Models. Journal of Petroleum Exploration and Production Technology, 2018, 8:1051. <https://doi.org/10.1007/s13202-018-0463-2>

Jing Lu, Fu Dai. Boundary Dominated Flow in Low Permeability Reservoir with Threshold Pressure Gradient. ARPN Journal of Engineering and Applied Sciences, Vol. 12, No. 23, 2017.

Education

2023.06-2024.12	Arizona State University	M.S. in Computer Science
2014.09-2016.06	Khalifa University	M.S. in Petroleum Engineering
2009.09-2012.06	China University of Petroleum (Beijing)	M.S. in Oil Gas Engineering
2005.09-2009.06	Tianjin University	B.S. in Science

Certifications & Training

2023 – Azure Data Engineering Certificate

2021 – Community Contributor, FATE Federated Learning Platform (WeBank, Shenzhen)

2020 – Huawei HCIE & HCIP Big Data Certificates

2018 – Reinforcement Learning, Coursera

2017 – Deep Learning, Coursera

2017 – Data Science and Big Data Analytics: Making Data-Driven Decisions, MIT